

ELITE MATHEMATICS

Pearson Edexcel International A Level

Pure 1

Adaptive Classified

Chapter 05 · Polynomials

Questions

Student book · questions with writing space

- 19 pages
- Past papers up to June 2026

Polynomials

Your classified route

6

questions

51

marks

3

families

7

variants

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CLASSIFIED BY IDEA

Move from recognition to method, then use the working space to complete the solution.

Polynomials

Families and variants

F01 • Factor and Solve Cubics

V01 Determine a Parameter, then Factor the Cubic

V02 Common Factor, then Quadratic Factor

F02 • Build a Polynomial from Roots and Graph Behaviour

V01 Distinct Roots and a Point

V02 Repeated Root from Touch Behaviour

V03 Intersection by Preserving Common Factors

F03 • Transfer Roots Through Transformations or Substitutions

V01 Input-transformation Root Map

V02 Substituted-variable Root Transfer with a Domain Filter

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Factor and Solve Cubics

WHAT THIS FAMILY ASKS

Determine any required coefficient, extract a common factor and factor the remaining quadratic to obtain every real root of a polynomial of degree at most three.

VARIANT ROUTE

V01 Determine a Parameter, then Factor the Cubic

V02 Common Factor, then Quadratic Factor

2 Complete Classified questions

16 total marks

2.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

$$f(x) = ax^3 + (6a + 8)x^2 - a^2x$$

where a is a positive constant.

Given $f(-1) = 32$

- (a) (i) show that the only possible value for a is 3

- (ii) Using $a = 3$ solve the equation

$$f(x) = 0$$

(5)

$$f(x) = ax^3 + (6a + 8)x^2 - a^2x$$

$$a = 3$$

(b) Hence find all real solutions of

(i) $3y + 26y^{\frac{2}{3}} - 9y^{\frac{1}{3}} = 0$

(ii) $3(9^{3z}) + 26(9^{2z}) - 9(9^z) = 0$

(5)

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4.

**In this question you must show detailed reasoning.
Solutions relying on calculator technology are not acceptable.**

$$f(x) = x^2(2x + 1) - 15x$$

(a) Solve

$$f(x) = 0$$

(4)

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$$f(x) = x^2(2x + 1) - 15x$$

$$f(x) = 0$$

(b) Hence solve

$$y^{\frac{4}{3}} \left(2y^{\frac{2}{3}} + 1 \right) - 15y^{\frac{2}{3}} = 0 \quad y > 0$$

giving your answer in simplified surd form.

(2)

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Build a Polynomial from Roots and Graph Behaviour

WHAT THIS FAMILY ASKS

Translate x-intercepts and touch/cross behaviour into factors with the correct multiplicities, determine a scale from another point, and exploit factor structure in later intersections.

VARIANT ROUTE

V01 Distinct Roots and a Point

V02 Repeated Root from Touch Behaviour

V03 Intersection by Preserving Common Factors

4 Complete Classified questions

35 total marks

8. The curve C_1 has equation

$$y = 3x^2 + 6x + 9$$

(a) Write $3x^2 + 6x + 9$ in the form

$$a(x + b)^2 + c$$

where a , b and c are constants to be found.

(3)

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8. The curve C_1 has equation

$$y = 3x^2 + 6x + 9$$

The point P is the minimum point of C_1

(b) Deduce the coordinates of P .

(1)

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8. The curve C_1 has equation

$$y = 3x^2 + 6x + 9$$

A different curve C_2 has equation

$$y = Ax^3 + Bx^2 + Cx + D$$

where A, B, C and D are constants.

Given that C_2

- passes through P
- intersects the x -axis at $-4, -2$ and 3

(c) find, making your method clear, the values of A , B , C and D .

(5)

4.

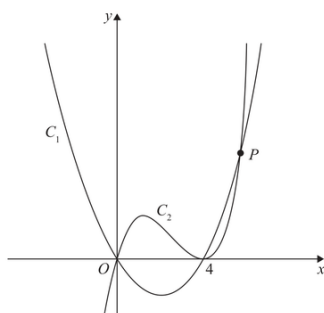


Figure 1

Figure 1 shows a sketch of part of the curves C_1 and C_2

Given that C_1

- has equation $y = f(x)$ where $f(x)$ is a quadratic function
- cuts the x -axis at the origin and at $x = 4$
- has a minimum turning point at $(2, -4.8)$

(a) find $f(x)$

(3)

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Given that C_2

- has equation $y = g(x)$ where $g(x)$ is a cubic function
- cuts the x -axis at the origin and meets the x -axis at $x = 4$
- passes through the point $(6, 7.2)$

(b) find $g(x)$

(3)

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Figure 3 shows a sketch of the curve C with equation $y = f(x)$ where $f(x)$ is a cubic function in x .

- cuts the x -axis at $(-2, 0)$ and cuts the y -axis at $(0, 10)$

- touches the x -axis at $(5, 0)$

(a) Deduce the roots of the equation

$$(i) \quad f\left(\frac{1}{3}x\right) = 0$$

(ii) $f(x - 3) = 0$

(2)

The curve C

- as shown in Figure 3.

(3)

The curve C

- cuts the x -axis at $(-2, 0)$ and cuts the y -axis at $(0, 10)$
- touches the x -axis at $(5, 0)$

The curve C intersects the straight line $y = 10(x + 2)$ at exactly three points.

(Solutions based entirely on calculator technology are not acceptable.)

(4)



The curve C

- Given that $f(x)$ is a cubic function,

(3)

The curve C

- touches the x -axis at $-\frac{3}{2}$
- cuts the x -axis at 4
- cuts the y -axis at -12

Given that $f(x)$ is a cubic function,

The line with equation $y = 2x - 12$ cuts C at $(0, -12)$ and at two further points, P and Q .

- (b) Use algebra to find the exact x coordinates of P and Q .
Give each answer in simplest form.

(Solutions relying entirely on calculator technology are not acceptable.)

(4)